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Vulnerability to environmental hazards

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For over 50 years, hazards researchers have focused on a series of fundamental questions: 1) what is the human occupancy of hazard zones? 2) How do people and societies respond to environmental hazards and what factors influence their choice of adjustments? 3) How do you mitigate the risk and impact of environmental hazards? Within the last decade or so, another question was added to our ongoing list of imponderables: are societies becoming more vulnerable to environmental hazards?

Vulnerability, broadly defined as the potential for loss, is an essential concept in hazards research and is central to the development of hazard mitigation strategies at the local, national and international level. Within the framework of the UN's International Decade of Natural Disaster Reduction (IDNDR), for example, vulnerability assessments are used to determine the potential damage and loss of life from extreme natural events. They are important as well in proposing hazard reduction alternatives where mitigation normally takes the form of structural (engineered) approaches to hazard reduction (Coburn and Spence, 1992; Clayton, 1994). Rarely mentioned are the underlying causes of increased social vulnerability to hazards or disaster events. Yet, vulnerability now forms the cornerstone of international efforts aimed at reversing the poverty, population, development and environmental degradation downwards spiral. This developmental-based approach to vulnerability reduction is best seen in Cuny's (1983) classic work on *Disasters and development*. Despite these noteworthy efforts, we still lack a common conceptualization of vulnerability.

This progress report reviews the recent research on vulnerability in the hazards field. This critical evaluation focuses on three thematic areas: the confusion and contradiction on the meaning of the term; measurement; and the causes of spatial outcomes associated with vulnerability studies. A new conceptual model of vulnerability is then proposed, the hazards of place, which clarifies many of the discrepancies found in the existing literature.

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I Fuzzy definitions and divergent themes

To begin the discussion we need to distinguish between the origins of vulnerability, for this is often the initial point of disagreement. While the broad definition of vulnerability infers a potential for loss, it never clearly articulates what type of loss and whose loss we are describing. There is personal or individual potential for, or sensitivity to, losses (or harm) that have both spatial and nonspatial domains. This is termed individual vulnerability. Social vulnerability includes the susceptibility of social groups or society at large to potential losses (structural and nonstructural) from hazard events and disasters. It has distinct spatial outcomes and varies over time. Finally, there is a potential for loss derived from the interaction of society with biophysical conditions which in turn affect the resilience of the environment to respond to the hazard or disaster as well as influencing the adaptation of society to such changing conditions. The biophysical vulnerability also has explicit spatial outcomes. For this report, I will focus more on those sources of vulnerability that produce explicit spatial outcomes, social and biophysical, and describe their origins and common usage within the social science hazards literature.

Vulnerability is frequently used in the risk, hazards and disasters literature (Gilbert, 1995; Hewitt, 1983; 1995), but is becoming more prominent in the areas of global change and environment and development studies (Liverman, 1990a; Dow, 1992; Dow and Downing, 1995). Within geography, the vulnerability concept has been in use for more than a decade with one of the best conceptualizations written in the early 1980s (Timmerman, 1981). Despite more than a decade's worth of collective research experience on the concept, vulnerability still means different things to different people (Table 1).

Many of the discrepancies in the meanings of vulnerability arise from different epistemological orientations (political ecology, human ecology, physical science, spatial analysis) and subsequent methodological practices. Also, there is considerable variation in the choice of hazards themselves (famine, floods, drought, seismic events, technology) and in the regions chosen for examination (developed versus developing countries). Fundamental conceptual differences exist, as well, visions that either focus research on the likelihood of exposure (biophysical/technological risk), the likelihood of adverse consequences (social vulnerability) or some combination of the two. The result is a confused lexicon of meanings and approaches to understanding vulnerability to environmental hazards. At the very least, some form of clarification or consistency in our use of vulnerability would be helpful if we are to advance our theoretical and practical understanding of how and why places and people are vulnerable to environmental hazards. At this juncture, one can find three distinct themes in vulnerability studies, although others have suggested alternative categorizations (Liverman, 1990a; Dow, 1992). They are vulnerability as risk/hazard exposure; vulnerability as social response; and vulnerability of places. While not mutually exclusive nor exhaustive, this typology helps to distinguish between the theoretical and methodological orientations of the current research.

1 Vulnerability as pre-existing condition

The first research theme examines the source (or potential exposure or risk) of biophysical or technological hazards. These studies are characterized by a focus on the distri-

Table 1 Selected definitions of vulnerability

Gabor and Griffith (1980)
Vulnerability is the threat (to hazardous materials) to which people are exposed (including chemical agents and the ecological situation of the communities and their level of emergency preparedness). Vulnerability is the risk context.

Timmerman (1981)
Vulnerability is the degree to which a system acts adversely to the occurrence of a hazardous event. The degree and quality of the adverse reaction are conditioned by a system's resilience (a measure of the system's capacity to absorb and recover from the event).

UNDRO (1982)
Vulnerability is the degree of loss to a given element or set of elements at risk resulting from the occurrence of a natural phenomenon of a given magnitude.

Susman et al. (1984)
Vulnerability is the degree to which different classes of society are differentially at risk.

Kates (1985)
Vulnerability is the 'capacity to suffer harm and react adversely'.

Pijawka and Radwan (1985)
Vulnerability is the threat or interaction between risk and preparedness. It is the degree to which hazardous materials threaten a particular population (risk) and the capacity of the community to reduce the risk or adverse consequences of hazardous materials releases.

Bogard (1989)
Vulnerability is operationally defined as the inability to take effective measures to insure against losses. When applied to individuals, vulnerability is a consequence of the impossibility or improbability of effective mitigation and is a function of our ability to detect the hazards.

Mitchell (1989)
Vulnerability is the potential for loss.

Liverman (1990a)
Distinguishes between vulnerability as a biophysical condition and vulnerability as defined by political, social and economic conditions of society. She argues for vulnerability in geographic space (where vulnerable people and places are located) and vulnerability in social space (who in that place is vulnerable).

Downing (1991b)
Vulnerability has three connotations: it refers to a consequence (e.g., famine) rather than a cause (e.g., drought); it implies an adverse consequence (e.g., maize yields are sensitive to drought; households are vulnerable to hunger); and it is a relative term that differentiates among socioeconomic groups or regions, rather than an absolute measure of deprivation.

Dow (1992)
Vulnerability is the differential capacity of groups and individuals to deal with hazards, based on their positions within physical and social worlds.

Smith (1992)
Risk from a specific hazard varies through time and according to changes in either (or both) physical exposure or human vulnerability (the breadth of social and economic tolerance available at the same site).

Continued.

Table 1 Continued

Alexander (1993)
Human vulnerability is a function of the costs and benefits of inhabiting areas at risk from natural disaster.

Cutter (1993)
Vulnerability is the likelihood that an individual or group will be exposed to and adversely affected by a hazard. It is the interaction of the hazards of place (risk and mitigation) with the social profile of communities.

Watts and Bohle (1993)
Vulnerability is defined in terms of exposure, capacity and potentiality. Accordingly, the prescriptive and normative response to vulnerability is to reduce exposure, enhance coping capacity, strengthen recovery potential and bolster damage control (i.e., minimize destructive consequences) via private and public means.

Blaikie et al. (1994)
By vulnerability we mean the characteristics of a person or group in terms of their capacity to anticipate, cope with, resist and recover from the impact of a natural hazard. It involves a combination of factors that determine the degree to which someone's life and livelihood are put at risk by a discrete and identifiable event in nature or in society.

Bohle et al. (1994)
Vulnerability is best defined as an aggregate measure of human welfare that integrates environmental, social, economic and political exposure to a range of potential harmful perturbations. Vulnerability is a multilayered and multidimensional social space defined by the determinate, political, economic and institutional capabilities of people in specific places at specific times.

Dow and Downing (1995)
Vulnerability is the differential susceptibility of circumstances contributing to vulnerability. Biophysical, demographic, economic, social and technological factors such as population ages, economic dependency, racism and age of infrastructure are some factors which have been examined in association with natural hazards.

bution of some hazardous condition, the human occupancy of this hazardous zone (e.g., floodplains, coastal areas, seismic zones), and the degree of loss (life and property) associated with the occurrence of a particular event (flood, hurricane, earthquake). Magnitude, duration, impact, frequency and rapidity of onset all characterize these exposure or biophysical vulnerability studies (Hewitt and Burton, 1971; Gabor and Griffith, 1980; Ambraseys and Jackson, 1981; Gabor and Pelanda, 1982; UNDRO, 1982; Pijawka and Radwan, 1985; Cutter and Solecki, 1989; Cutter and Tiefenbacher, 1991; Heyman *et al.*, 1991; Burton *et al.*, 1992; Haque and Blair, 1992; Quarantelli, 1992; Swearingen, 1992; Alexander, 1993; Parrish *et al.*, 1993). A subset of studies examines the distribution of structural losses and vulnerability reduction in the built environment associated with natural disasters events (Harlan, 1988; Coburn and Spence, 1992; Corsanego *et al.*, 1993; Tavakoli and Tavakoli, 1993; Solway, 1994).

2 Vulnerability as tempered response

The second group of vulnerability studies focuses on coping responses including societal resistance and resilience to hazards. The nature of the hazardous event or con-

dition is usually taken as a given, or at the very minimum viewed as a social construct not a biophysical condition. Much of the research examines chronic disturbances such as drought, famine, hunger, climate change or environmental change (Susman *et al.*, 1984; Bogard, 1989; Chambers, 1989; Anderson and Woodrow, 1991; Downing, 1991a; 1991b; 1992; Watts and Bohle, 1993; Blaikie *et al.*, 1994; Bohle *et al.*, 1994; Chen, 1994; Yarnal, 1994). This perspective highlights the social construction of vulnerability, a condition rooted in historical, cultural social and economic processes that impinge on the individual's or society's ability to cope with disasters and adequately respond to them.

3 Vulnerability as hazard of place

While vulnerability as potential exposure or social response pervades the literature, a third direction is emerging that combines elements of the two, but which is inherently more geographically centred. In this perspective, vulnerability is conceived as both a biophysical risk as well as a social response, but within a specific areal or geographic domain. This can be geographic space, where vulnerable people and places are located, or social space, who in those places are most vulnerable. Recently, a number of researchers have used this integrative approach in a wide array of spatial contexts or places (local, regional, national) (Blaikie and Brookfield, 1987; Wilhite and Easterling, 1987; Mitchell *et al.*, 1989; Lewis, 1987; 1990; Liverman, 1986; 1990b; Palm and Hodgson, 1992a; Degg, 1993; Longhurst, 1995).

II Measurement and assessment techniques

While the three theoretical streams run parallel courses, the methodological approaches to vulnerability are highly varied. Since there is little consistency in the definitions of vulnerability, we would expect it to be quite difficult to operationalize the concept using specific variables or indicators. Remarkably, the measurement task is somewhat easier than the definitional one. Variables related to exposure normally include proximity to the source of threat, incident frequency or probability, magnitude, duration or spatial impact. Social impact and response are often measured by threats to lifelines or infrastructure to support basic needs (Gasser and Snitofsky, 1990; Platt, 1991), special needs populations (children, elderly, infirm), poverty/wealth indicators, gender, race and so forth. Often, the vulnerability indicators are single variables but multidimensional factors such as food aid, institutional development, social relations or political power have been employed as well (Watts and Bohle, 1993; Blaikie *et al.*, 1994).

Geographic scale poses another measurement difficulty for vulnerability with applications ranging from local to global scales. Detailed vulnerability assessments are conducted most often at the micro or local level (Colten, 1986; 1991; OAS, 1991) with an occasional regional application (Downing, 1992; Lowry *et al.*, 1995). Methodological decisions often mean sacrificing localized detailed case-study approaches for more broadly based patterns and distributions. The choice of areal unit within local studies also ranges from individual household level to geopolitical census units.

The time dimension poses other problematic issues for vulnerability studies. There are relatively few historical treatments of vulnerability with the exception of Colten's (1988; 1993) work on hazardous waste in Illinois. The temporal context of vulnerability

is crucial as alluded to elsewhere (Cutter, 1993), yet the temporal dimension remains one of the least studied aspects of vulnerability. It is unclear, at this time, why this is so.

Lastly, the techniques used in vulnerability studies are diverse as well. The analytical approaches closely follow research paradigms and run the gamut from historical narratives (Colten, 1991), contextual analyses (Mitchell *et al.*, 1989), case studies (Liverman, 1990b), to statistical analyses, GIS and mapping techniques (McMaster and Johnson, 1987; McMaster, 1988; Burke, 1993; Parrish *et al.*, 1993; von Braun, 1993; Wadge *et al.*, 1993; Hepner and Finco, 1995).

III Causal linkages and spatial outcomes

The vulnerability literature is bifurcated when it comes to an understanding of causes of vulnerability, which is not a surprise considering the different theoretical orientations. The vast majority of vulnerability studies take a political–economic perspective and suggest a causal structure that concentrates on the differential social impacts and abilities to cope with the crisis at hand. Causal explanations of the vulnerability arise from underlying social conditions that often are quite remote from the initiating event itself (e.g., a natural event, technological failure). Others contend that vulnerability is locationally driven and is a function of the proximity to the source of the threat.

1 Causal structure

The causes of social vulnerability are explained by the underlying social conditions that are often quite remote from the initiating hazard or disaster event. For example, Watts and Bohle (1993) propose one such explanatory model in their analysis of hunger and famine. In their view, vulnerability is defined by three processes – entitlement (or economic capability), empowerment (political/social power) and political economy (historical/structural class-based patterns of social reproduction). They suggest that the intersection of these tripartite processes produces the social space of vulnerability. This model is then applied to a comparative study of south Asia and sub-Saharan Africa to examine famine vulnerability over space and time. In a subsequent analysis (Bohle *et al.*, 1994), the model is used to explain the social context of hunger and famine and vulnerability to climate change in Zimbabwe.

In another formulation, Blaikie *et al.* (1994) examine causality using two alternative models: the pressure and release model (PAR) and the access to resources model. The PAR model provides an explanation of the relationship between processes that give rise to 'unsafe conditions' (e.g., exposure) and their intersection with some type of hazard or disaster event thus creating a form of social vulnerability. The focus is clearly on the dynamic pressures and underlying driving forces that give rise to vulnerability in the first place. The access to resources model is a more refined explanation of the role of political and economic forces as the root cause of the unsafe conditions.

2 Vulnerability/exposure assessments

Another approach to understanding causality found in the literature explains the outcome by modelling the potential exposure to hazards. The assumption, of course, is that vulnerability is primarily a function of the proximity to the source of the risk or

hazard in question (Alexander, 1993). A simple mapping of the biophysical risk should result in a simplistic delineation of the likely exposure or biophysical vulnerability. Palm and Hodgson's (1992a) work on seismic mapping in relation to earthquake insurance purchase is a good example of this approach.

Within the hazards management community, vulnerability assessments are extensively used to delineate, characterize and assess the hazard with an eye towards the development of some mitigation strategy. The USFEMA's floodplain maps are one example of simple vulnerability assessments. The OAS (1991) also developed a series of multihazard maps that incorporate vulnerability assessments into their preimpact planning and mitigation efforts. These assessments include human populations, critical facilities and lifelines, economic production facilities, and differences in vulnerability among economic sectors. All these are mapped to illustrate both the biophysical/technological risk as well as the impact of this risk on these vulnerable indicators (population, economic activity, lifelines).

More recently, governmental agencies are using computer-assisted emergency information systems that incorporate some vulnerability indicators into them. For example, the USNOAA's ALOHA (Areal Locations of Hazardous Atmospheres) provides a plume dispersion model designed for airborne toxic releases to ascertain the risk to populations from potential chemical exposure. The USEPA's CAMEO (Computer-Aided Management of Emergency Operations) focuses more on preimpact planning and during-hazard event responses, again utilizing exposure indicators and populations at risk. The same is true of the National Response Team's manual for hazardous material spills (NRT, 1987). Included in these documents are potential scenarios on likely exposures and some indicators of demographic distributions and critical facilities, all indicators of social and biophysical vulnerability. Commercial emergency information systems vendors also provide software for exposure and vulnerability assessments from a wide range of hazards. This increased use of geographic information systems (GIS) and related technologies by researchers, managers, governmental agencies and nongovernmental groups interested in hazards and disasters is prompting us to reconsider our understanding of vulnerability, both in theoretical and applied ways.

IV Towards a new synthesis: the hazards of place revisited

In struggling with these issues for many years, it became necessary to clarify my own thinking and interest in vulnerability. Largely working within the North American context and with a category of hazards that arose from the interaction between nature, society and technology, it was clear to me that many of the existing theoretical constructs were either too limiting or too diffuse to be of practical use to the problems I examined. Out of a sense of intellectual continuity, I returned to the original work of Hewitt and Burton on the hazardousness of places (1971) where they attempted a multihazard mapping exercise to delineate a regional ecology of natural hazards events. Borrowing from them, the initial hazards of place model was proposed (Cutter and Solecki, 1989), and then tested with other hazards (Solecki, 1990; Tiefenbacher, 1992; Cutter, 1993).

The model of vulnerability in Figure 1 is the culmination of this journey coupled with inputs from the recent literature. It is exploratory in nature, but serves as a useful heuristic in understanding the diverse elements that contribute to our understanding

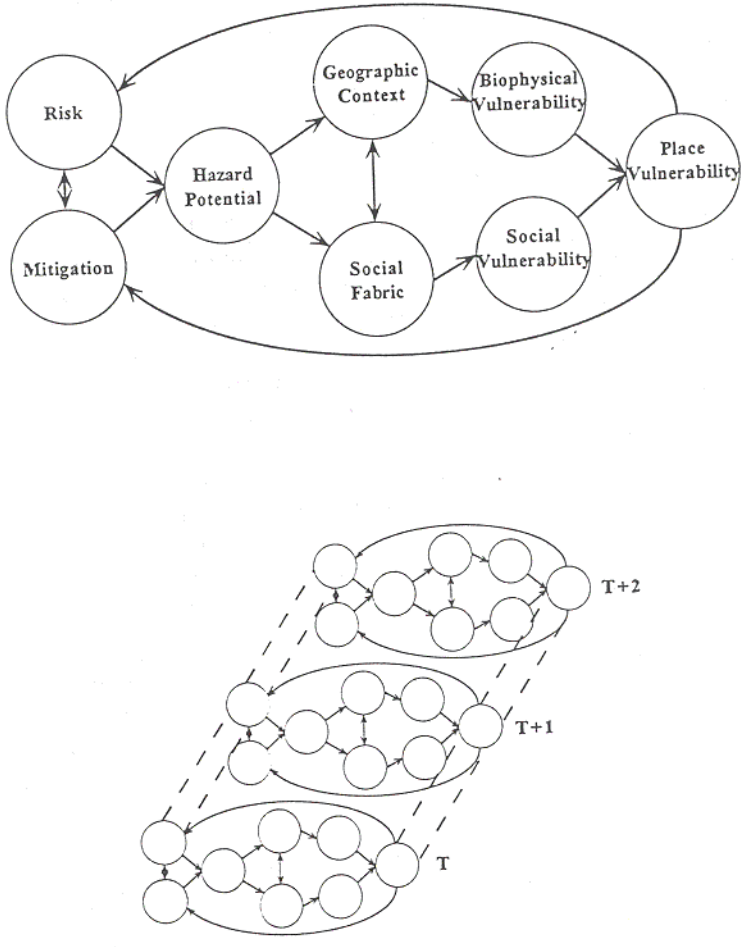


Figure 1 The hazards of place model of vulnerability. The various elements that constitute vulnerability interact to produce the vulnerability of specific places and the people who live there (top). This vulnerability can change over time (bottom) based on changes in the risk, mitigation and contexts within which environmental hazards occur

of the vulnerability of places. There is an explicit focus on locality within this conceptual framework, for it is place that forms the fundamental unit of analysis for any geographer. The model is simplistic in its presentation, yet we must understand that all the parameters change over time (as shown on the bottom half of the diagram). Also, each of the major indicators consists of a series of nested or contextual arrangements that influence it, which could easily be expanded but are not in this review.

Risk is the likelihood of occurrence (or probability) of the hazard. Risk has two domains: it includes the potential sources of risk (industrial, flooding, transportation) and the contextual nature of the risk itself (high consequence, low consequence). The second domain is a simple probabilistic estimate based on the frequency of occurrence (100-year flood, 5% risk). Risks combine with mitigation (efforts to reduce risks such as planning, prior experience) to create an overall hazard potential. Risks can be attenu-

ated with good mitigation or they can be amplified by a poor or nonexistent mitigation practices. The hazard potential is filtered through the social fabric of society (socioeconomic indicators, cognition of risk, individual/societal ability to respond) to determine the overall social vulnerability of the place. The hazard potential is also filtered through its geographic context (site and situation, proximity) to determine biophysical/technological vulnerability. It is the intersection and interaction of both the social vulnerability and biophysical/technological vulnerability that create the vulnerability of places. The place vulnerability provides a feedback loop to both the risk and mitigation, which in turn further reduces or enhances both risk and mitigation.

The hazards of place model is a useful construct for examining vulnerability and incorporates the range of theoretical and conceptual issues outlined in this report. As was illustrated, vulnerability was traditionally viewed as either a pre-existing condition or potential exposure to a risk (biophysical) or as a social condition predisposing some response to an environmental threat (social vulnerability). The merger of these permits an examination of both the biophysical and social vulnerabilities as they affect places. The hazards of place model can facilitate a single or multihazards approach with differing hazard characteristics (chronic and acute), contrasting contexts (social, political, economic) and diverse methodological approaches (historical, GIS, descriptive, empirical). While empirical verification is still at a nascent stage, the simplicity of the hazards of place model suggests wide application in our comparative understanding of vulnerable places and the people who live there.

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